QUESTION-TO-SLIDE MAPPING

SECTION B - LECTURE 01

Climate Introduction: Climatic Study & Bioclimatic Architecture

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	1.0 - Climate Introduction Billah Sir (8 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map the introductory lecture only to questions it can genuinely support: course/professional value, why climatic conditions create building variation, passive-climatic framing and the bioclimatic idea. Global mechanisms, tropical-climate details and climatic measurement are left to later Section B lectures.

RESULT	Section B Lecture 01 is a foundation lecture rather than a long-answer engine. It gives Exact coverage for the 2016 and 2017 questions on the importance/professional use of climatic study. Later passive-control, climatic-factor and traditional-form questions are only Partial because their detailed mechanisms belong to subsequent lectures.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

Lecture 01 establishes why climatic study matters in building design. It states that buildings should fit their climate, links building climatology to comfort and energy-saving structural design, emphasizes passive protection and natural resources, explains why climate creates regional building variation, and introduces bioclimatic architecture as a relationship between living organisms and architecture. It is best used as the conceptual opening for later Section B answers.

8	8	2	0	6	85
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
2-3	Course scope: climate, elements, comfort, passive control, site/ecology	Supports many later questions	Orientation only; not enough detail for the later mechanism questions.
5	Benefits of climatic study: comfort, energy, protection, natural resources	2016 Q1(a); 2017 Q1(a)	Highest priority; directly answers the early course-value questions.
6	Why building forms vary with climate	2018/2024 design-response questions support	Useful conceptual bridge from climate to form, requirement, equipment and method.
7	Bioclimatic architecture	Supports passive/climate-responsive design answers	Key definition/concept; not a full tropical-design chapter.
8	Closing	None	No study time needed.

Priority note

Master Slides 5-7. For a 10-15 mark “importance/use” question, write: Comfort -> Energy saving -> Protection -> Natural resources -> Regional/form response. Use Slides 2-3 only as a course-scope checklist.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q1(a)	10	1	Figure out the importance of climate study for a BECM graduate.	Exact	3, 5-6	The lecture directly gives benefits of climatic study and connects them to building design decisions.
2017	Q1(a)	15	2	Identify areas where a BECM professional can utilize learning from Climate & Architectural Design.	Exact	2-6	Course scope, comfort/energy benefits, passive protection, natural-resource use and design variation directly support the answer.
2017	Q2(a)	15	2	Why is passive control in tropical climate more critical than in sub-tropical/polar climates?	Partial	3, 5	Passive control is emphasized, but tropical-climate reasons and comparison are not developed in Lecture 01.
2018	Q5(c)	10	3	Why are tropical climates of maximum importance to a built-form designer?	Partial	3, 5-6	The lecture explains why climate matters to design generally; tropical-specific characteristics belong to Lecture 04.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2022	Q1(a)	5	8	“Passive control in tropical climate is more critical...” justify.	Partial	3, 5	The passive-control principle is present; tropical-specific justification is not.
2023	Q1(d)	5	10	“Passive control in tropical climate is more critical...” justify your opinion.	Partial	3, 5	Same partial support as 2022; use the tropical-climate lecture for a complete answer.
2024	Q4(b)	13	13	What climatic factors affect building design and how do they influence it?	Partial	2-3, 6	The lecture shows that climate affects arrangement, requirements, equipment and method, but does not enumerate/detail all climatic elements.
2024	Q8(a)	12	15	Identify climates from two traditional built forms and analyze components, surroundings and orientation.	Partial	6-7	Building variation and bioclimatic principles support interpretation, but climate identification and detailed response come from later tropical-design lectures.

Coverage key	Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

This introductory deck generates fewer direct final questions than the later Section B mechanism lectures. Its value is that the same core framing can open several answers: climate affects building form, passive design precedes mechanical correction, and design should exploit environmental resources.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Importance / professional use of climatic study	2016, 2017	3, 5-6	Only fully direct repeated family; easy structured theory marks.
A	Passive climatic control / climate-responsive importance	2017, 2022, 2023 (+2018 related)	3, 5	Good opening logic, but tropical-specific proof is in Lecture 04.
A	Climate affects building form / requirements / equipment / method	2018, 2024 related	6	Useful design-significance framework.
B	Bioclimatic architecture / nature-responsive design	Conceptual support	7	Important definition that strengthens later design answers.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
“importance of climate study / professional use”	3, 5-6	Use Comfort - Energy - Protection - Natural resources - Regional/form response.
“why buildings vary from region to region”	6	Climate changes arrangement, requirements, equipment selection and building method/formation.
“bioclimatic architecture”	7	Relationship between living organisms and architecture; nature-inspired design optimizing/using the environment.
“passive control in tropical climate”	3, 5	Use only as introduction; complete the tropical-specific reasons from Lecture 04.
“climatic factors affecting design”	2-3, 6	Use as broad workflow, then move to Elements of Climate for the actual factor list.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
1	Course/lecture identity	None	Skip after recognizing lecture.
2	Syllabus map: global factors, elements, climate classification, site, ecology	Later Section B roadmap	Use only to know where complete answers live.
3	External/internal climate, building as climatic modifier, comfort/energy via passive design	2016 Q1(a); 2017 Q1(a); passive-control questions partial	Memorize the one-sentence course purpose.
4	Reference-book slide	None	No exam content to memorize.
5	Benefits of Climatic Study	2016 Q1(a); 2017 Q1(a); passive questions support	Highest-priority slide; memorize four benefits.
6	Climate -> building variation / arrangement / requirements / equipment / method	2018/2024 related design questions	Turn into a cause-effect flow diagram.
7	Bioclimatic architecture definition + environmental optimization	Passive/climate-responsive answers support	Memorize definition and one application sentence.
8	Closing	None	Skip.

6. Coverage Gaps & Diagnostic Exclusions

Lecture 01 deliberately introduces the course. Do not stretch its broad statements into complete answers for later technical questions.

Gap	Affected questions	Action
Global radiation / Earth-Sun / wind mechanisms	2016-2024 global-climate long questions	Use Section B Lecture 02.
Tropical climate classification and passive-control justification	2017, 2018, 2022, 2023, 2024 tropical questions	Use Section B Lecture 04 for climate characteristics and design response.
Temperature, humidity, precipitation, driving rain, wind measurement	Elements/measurement questions	Use Section B Lectures 03 and 03.1 separately.
Regional/site/urban climate and local deviations	Site-climate questions	Use Section B Lecture 05.

Mapping discipline	The deck explicitly states the climatic-study benefits and bioclimatic definition, but it does not supply the detailed mechanisms of later lectures. Therefore later questions are marked Partial even when Lecture 01 provides a useful introduction.

7. Recommended Study Plan

Treat this as a foundation note: memorize the high-yield statements, then use it to open answers from the later Section B lectures. Do not spend equal time on every slide.

Step	Study block	Slides	Practice immediately
1	Benefits of climatic study	5	2016 Q1(a), 2017 Q1(a).
2	Climate-driven building variation	6	Write a 4-arrow cause-effect flow from climate to design.
3	Bioclimatic architecture	7	Prepare a 3-5 line definition + one design implication.
4	Course purpose / passive-control framing	2-3	Use as introduction for 2017/2022/2023 passive-control questions.
5	Move to later lectures	2 roadmap	Lecture 02 global factors; 03/03.1 elements; 04 tropical; 05 site; 06 thermal comfort.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Benefits: Comfort - Energy - Protection - Natural resources	Write one design example after each benefit.
A	Climate -> building variation flow	Arrangement -> requirement -> equipment -> method/form.
A	Bioclimatic definition	Relationship between living organisms and architecture + environmental optimization.
B	Course-scope sentence	Building acts as a climatic modifier to provide comfort/energy saving through passive design.

Bottom line	Section B Lecture 01 is small but important: it gives the clean conceptual language for why the course matters. Secure the two direct early-paper questions, then use the lecture as an introduction - not a substitute - for the detailed Section B mechanisms that follow.